# EVOLUTIONARY DISCOVERY OF REINFORCEMENT LEARNING ALGORITHMS VIA LARGE LANGUAGE MODELS

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# INTRODUCTION

Framing the problem

# WHAT RL IS USED FOR

### Reasoning LLMs

o1, DeepSeek-R1 — RL makes them think

### LLM alignment

RLHF, DPO, GRPO, Constitutional AI

### Robotics & autonomy

$\pi_0$, OpenVLA, Atlas, Waymo, AlphaStar

### Science & operations

AlphaDev, AlphaTensor, chip design, data-center cooling

# THE PROBLEM WITH RL & ES

## Reinforcement Learning

- A handful of hand-crafted update rules
- Brittle to hyperparameters and reward shaping
- Sample inefficient, slow to converge
- New algorithms = years of human effort

## Evolutionary Search

- Flexible, gradient-free, parallelizable
- But trapped in hand-designed search spaces
- DSLs constrain what can be expressed
- Random mutation wastes compute

# THE CLAIM

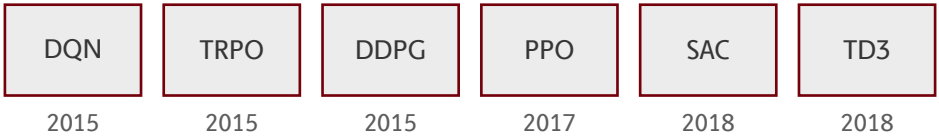
First paper to evolve entire RL update rules as executable Python code using an LLM as the variation operator.

The idea. A candidate algorithm is a Python `compute_loss` function. The LLM mutates and recombines code; real training runs score it. Actor–critic, value functions, and Bellman updates are banned in the prompt — so only genuinely new mechanisms survive.

Lineage. AutoRL had tried this with hand-crafted DSLs (Co-Reyes '21) or black-box neural nets (LPG, DiscoRL) — both boxed the search space in. LLM-code-evolution (FunSearch, AlphaEvolve) hit big math wins, and REvolve evolved rewards — but never the algorithm itself.

My take. Impressive proof-of-concept — but compute cost, narrow benchmarks, and the strong possibility that the LLM is remembering more than it's discovering leave real problems on the table.

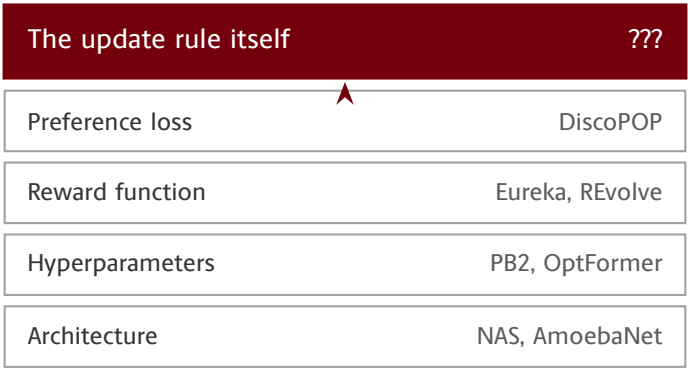
# MODERN RL RESTS ON 5 RULES




A decade of progress — hand-designed update rules.

Each took months of PhD-level effort. Each became dogma.

# AUTOML CLIMBS THE LADDER




The loss function was the last frontier.

# BACKGROUND

Two lineages converge on one idea

# LINEAGE A: DISCOVER AN RL ALGORITHM

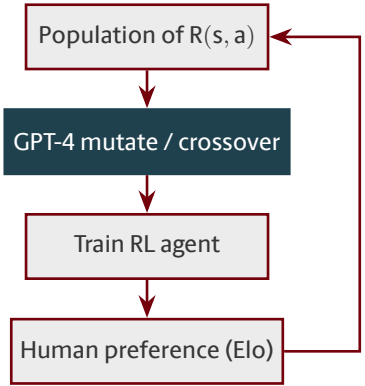

All constrain the search space by hand.

# LINEAGE B: LLMS MUTATE CODE

| ELM ’22 | FunSearch ’24 | AlphaEvolve ’25 | EvoTune ’25 |
|---|---|---|---|
| LLMs as mutators | cap-set, bin-packing | full files, Strassen beaten | proposer fine-tuning |

LLMs are intelligent variation operators over code.

# PARENT WORK: REVOLVE (2024)




What REvolve evolves

- Executable Python reward functions
- Island populations, LLM variation
- Tested on driving, humanoid, dexterous hands

The move in this paper.
Evolve the loss function instead of the reward.

# THE GAP THIS PAPER FILLS

| What gets evolved | | Representative method |
|---|---|---|
| Architecture | ✓ | NAS, AmoebaNet |
| Hyperparameters | ✓ | PB2, OptFormer |
| Reward function | ✓ | REvolve, Eureka |
| Preference loss (align) | ✓ | DiscoPOP |
| Update rule (DAG) | ✓ | Co-Reyes 2021 |
| Update rule (black-box NN) | ✓ | LPG, DiscoRL |
| Update rule (executable Python, LLM-evolved) | ★ | Sygkounas et al. 2026 |

Everything above is context. The bottom row is the thesis.

# METHOD

How the framework works

# THE BIG PICTURE

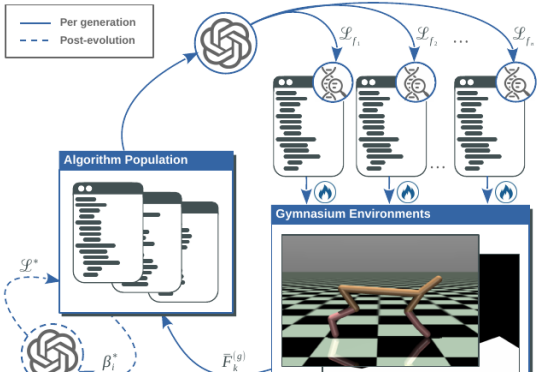



Figure 1 from the paper.

Per generation

- LLM proposes candidate loss functions
- Each is trained end-to-end on 5 Gym envs
- Fitness drives selection

Post-evolution

- LLM proposes hyperparameter ranges
- Sweep finds the refined rule $\mathcal{L}^\star$

# A CANDIDATE IS A CHUNK OF PYTHON

Parent

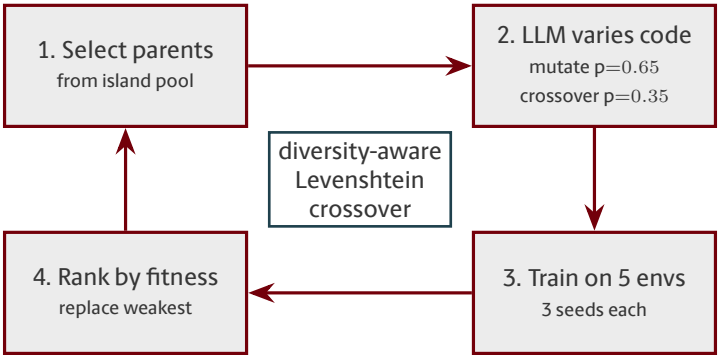
```
def compute_loss(theta, xi, D):
  preds = model(D.s, theta)
  targets = rollout(model, D.s)
  L = mse(preds, targets)
  return L
```

After LLM mutation

```
def compute_loss(theta, xi, D):
  preds = model(D.s, theta)
  conf = xi.confidence(D.s)
  targets = rollout(model, D.s)
  L = (conf * mse(preds, targets)).mean()
  return L
```

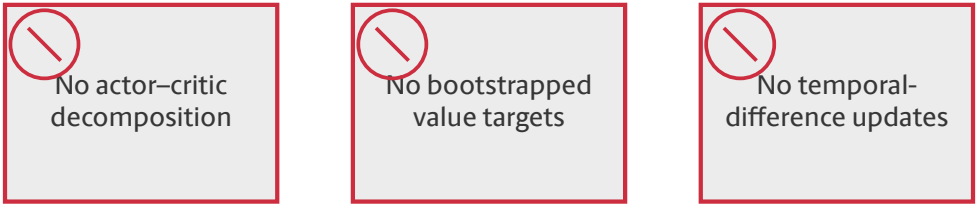
The LLM edits Python. That is the entire search space.

# THE EVOLUTIONARY LOOP




Island model · 9 individuals per island · 24 new candidates per generation.

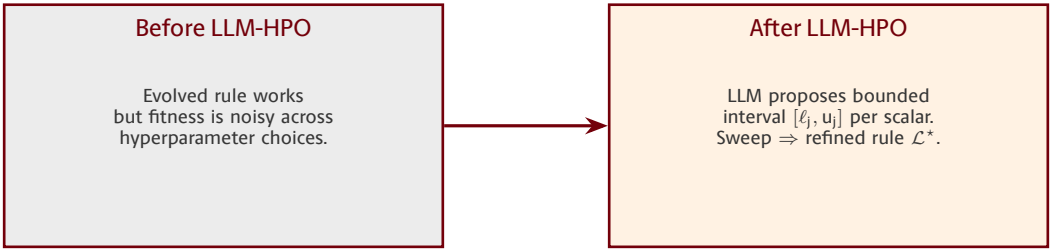
# WHAT THE LLM IS TOLD NOT TO DO




Constitutional prompt constraints $\Rightarrow$ forces novel mechanisms.

Evolution cannot re-invent DQN; it has to find something else.

# LLM-GUIDED HPO PASS




Policy arch, optimizer, rollout loop stay frozen. Only internal scalars move.

# RESULTS

What evolution discovered

# THE TWO ALGORITHMS IT INVENTED

**CG-FPD**

Confidence-Guided Forward Policy Distillation

Distill short-horizon plans from a learned latent dynamics model. Planning acts as the supervised teacher signal.

No value function. No policy gradient. No Bellman.

**DF-CWP-CP**

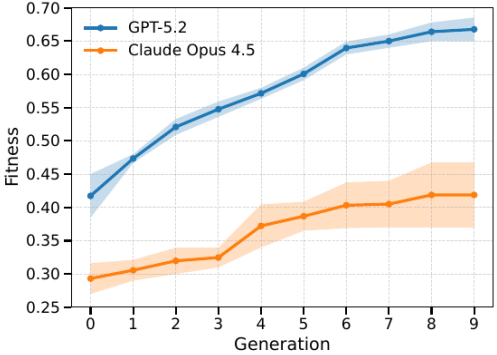
Differentiable Forward Confidence-Weighted Planning with Controllability Prior

Differentiable world-model rollouts + confidence-weighted desirability + controllability prior.

Also avoids every canonical RL mechanism.

Evolution rediscovered model-based planning — from scratch.

# THE PROPOSER LLM MATTERS. A LOT.




- GPT-5.2 climbs to 0.67
- Claude 4.5 Opus caps at 0.42
- Both monotone; GPT-5.2 is sharper on every generation

Same framework, same envs, same seeds — just swap the LLM.

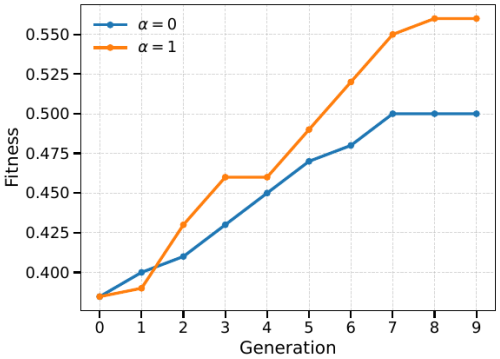
# RESULTS MATCH PPO AND SAC

| Environment | PPO | SAC | CG-FPD | DF-CWP-CP |
|---|---|---|---|---|
| CartPole | 500.0 | — | 500.0 | 500.0 |
| LunarLander | 246.6 | — | 241.2 | 260.6 |
| MountainCar | $-128.1$ | — | $-105.8$ | $-108.6$ |
| Acrobot | $-63.5$ | — | $-90.6$ | $-78.6$ |
| HalfCheetah | 1579.1 | 4988.5 | 2407.9 | 2103.8 |
| Swimmer | 95.0 | 87.8 | 247.5 | 219.8 |
| Walker2d | 3163.2 | 4595.3 | 1603.8 | 1297.6 |
| InvertedPendulum | 1000.0 | 1000.0 | 1000.0 | 1000.0 |

Competitive with hand-designed baselines across 10 environments.

Evolved on 5 training envs; remaining 5 are generalization tests.

# ABLATION: DIVERSITY WEIGHT




Levenshtein weight $\alpha$

- $\alpha = 0$: no structural regularization $\Rightarrow 0.50$
- $\alpha = 1$: full regularization $\Rightarrow 0.56$
- $\alpha = 0.5$: $0.65$

Some code-similarity nudge helps; too much or too little hurts.

# DISCUSSION

Critique and open questions

# STRENGTHS VS. WEAKNESSES

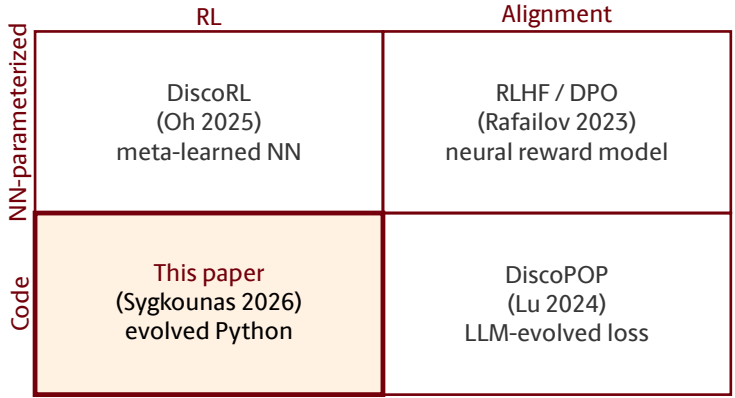
## Strengths

- First full update-rule discovery via LLM
- Genuinely novel mechanisms (no value, no PG)
- Clean extension of REvolve's machinery
- LLM-HPO handles fitness fragility cleanly

## Weaknesses

- 30 h per candidate on 4×A100
- Narrow benchmarks (CartPole / MuJoCo only)
- Is GPT-5.2 discovering or remembering?
- Only 2 evolutionary seeds per LLM
- No head-to-head vs. DiscoRL
- Constitutional constraints never ablated

# WHERE THIS PAPER SITS IN THE FIELD




The real contest — evolved source code vs. meta-gradient for RL.

# OPEN QUESTIONS

- Surrogate fitness to break the 30 h/candidate wall.
- Lift the constraints — let evolution decide whether to re-use actor–critic.
- Scale to Atari and robotics — generalization is still untested.
- Curriculum over environments — connect to POET, XLand.
- Self-improving proposer — EvoTune-style DPO on the evolved database.

Takeaway. REvolve showed LLMs can evolve rewards. This paper shows they can evolve the entire algorithm. The open question is whether what they discover is genuinely new — or just a good remix of what the model already saw.

# THANK YOU!

Questions & Discussion

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